Multisource Joint Representation Learning Fusion Classification for Remote Sensing Images

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***Abstract*— Multisource remote sensing images provide complementary multidimensional information for reliable and accurate classification. However, gaps in imaging mechanisms result in heterogeneity between multiple source images. During fusion, this heterogeneity causes the generated multisource representations to be redundant and ignore discriminative unisource information, which significantly hampers the fusion classification performance. To address this challenge, we introduce a novel multisource joint representation learning method for remote sensing image fusion classification, termed multisource information bottleneck fusion network (MIBF-Net). Based on the information bottleneck principle, MIBF-Net uses mutual information constraints to effectively integrate multisource information, generating a comprehensive and nonredundant multisource representation. Specifically, MIBF-Net first introduces an attribution-driven noise adaptation layer to dynamically balance the speed of feature learning across sources for extracting discriminative unisource intrinsic information. Furthermore, a cross-source relationship encoding (CRE) module is designed to fully explore cross-source complex dependencies for enhancing the richness of fused representations. Finally, we design an information bottleneck fusion (IB-Fusion) module to fuse unisource semantic information and cross-source information while reducing redundancy. In particular, we use variational inference techniques to effectively address the mutual information optimization problem and provide theoretical derivations. Extensive experimental results on three heterogeneous multisource remote sensing data benchmarks show that the model significantly outperforms the state-of-the-art methods.**

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***Index Terms*— Fusion classification, information bottleneck, multisource remote sensing.**

I. Introduction

REMOTE sensing image classification is widely used in many fields, such as earth observation, urban development, and forest management [1], [2], [3]. The continuous evolution of satellite sensor technology provides diverse remote sensing images [4], [5], [6]. For example, hyperspectral images (HSIs) offer rich spectral information but have low spatial resolution and are susceptible to cloud cover. Synthetic aperture radar (SAR) is able to acquire and penetrate data around the clock and provide geometric information on amplitude and phase. Light detection and ranging (LiDAR) collects depth and intensity information and measures the height of objects on the Earth's surface.

Multisource remote sensing images capture information about the same scene but reflect different characteristics due to their different imaging mechanisms. These heterogeneous characteristics provide distinct perspectives and properties that are essential for a comprehensive and accurate interpretation of remote sensing scenes. The joint use of these multisource remote sensing images allows combining the advantages of each data source and significantly improves the classification accuracy [7], [8], [9]. Therefore, multisource remote sensing image fusion classification has obtained extensive attention in recent years.

The most critical task in multisource remote sensing image classification is properly extracting and effectively fusing these heterogeneous features. For better classification, it is necessary to learn a proper multisource representation that usually satisfies two conditions: 1) provide sufficient information to make an accurate prediction and 2) avoid containing irrelevant information for the prediction. The existing multisource remote sensing image fusion classification methods can be broadly classified into two categories: traditional methods and deep-learning-based methods. The traditional methods [10], [11], [12], [13], [14] rely on manually designed feature representations using morphological operators and subspace learning, resulting in constrained performance and significant dependence on parameter settings [15]. The deep-learning-based methods [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34]

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are able to automatically learn and capture complex features in the data, achieving higher classification accuracy.

Despite the significant advancements achieved by deep-learning-based approaches, there still exist open challenges that deserve further research.

1) *Heterogeneity Causes Optimization Imbalance Problems:* In a two-branch classification framework, the learning speed of the feature extraction may vary for different source data, resulting in underoptimized features for the slower learning data sources [35].
2) *Simple Linear Projection Insufficiently Captures Correlation Information Between Multisource Data:* The high-order nonlinear relationship exists between heterogeneous features [36]. Ignoring this complex relationship leads to inaccurate analysis and interpretation of multisource data.
3) *Multisource Representation Contains Redundant Information:* The multisource data fusion needs to consider both the unisource feature information and the cross-source interaction relationships. However, this representation tends to be redundant, resulting in confusion in prediction and an increased risk of overfitting [37].

Considering the above problems, this article develops a multisource information bottleneck fusion network (MIBF-Net) for multisource remote sensing image classification. The MIBF-Net aims to integrate unisource intrinsic information and the cross-source potential relationships into an integrated representation. First, MIBF-Net performs adequate feature extraction for each data source. Second, MIBF-Net models the relationships between multiple source features. Finally, MIBF-Net encodes the unisource intrinsic and cross-source associative information into a high-quality multisource representation. We evaluated MIBF-Net on three heterogeneous datasets. The experimental results demonstrate that MIBF-Net significantly outperforms the state-of-the-art methods in improving the classification accuracy.

The main contributions of this work are presented as follows.

1) A novel multisource joint representation learning model MIBF-Net is developed for heterogeneous remote sensing image fusion classification. MIBF-Net is able to fully mine the unisource intrinsic information and cross-source complex dependencies, encouraging a comprehensive and nonredundant multisource representation.
2) To address the optimization imbalance problem, an attribution-driven noise adaptive layer (AD-NAL) is designed for features' extraction. The AD-NAL precisely controls the optimization process of each source feature extraction which can help adequately extract unisource features. Therefore, the problem of unisource feature underoptimization is alleviated.
3) A cross-source relationship encoding (CRE) module is proposed to capture complex relationships among heterogeneous multisource image features. By dynamically learning the importance of each feature and capturing higher order interactions, the CRE module enhances the capability to explore nonlinear relationships and feature interactions patterns within the heterogeneous multisource data.
4) An information bottleneck fusion module (IB-Fusion) is designed to adaptively fuse multisource information into a high-quality representation. This module uses information constraints to selectively retain the most relevant and discriminative information while discarding redundant and irrelevant ones. Therefore, a compact and effective multisource representation is generated for classification.

The rest of this article is organized as follows. Section II first introduces the methods related to multisource remote sensing image fusion classification. Section III describes the proposed method in this study. In Section IV, the experiments and results are described, and the experimental results obtained from different methods are compared and analyzed. Finally, this article is summarized in Section V.

II. Related Work

Over the past decades, a wide variety of multisource remote sensing image fusion classification methods have emerged. These methods can generally be grouped into two main categories: traditional methods and deep-learning-based methods.

A. Traditional Methods

For traditional methods, morphological operators and subspace learning stand as representative techniques [15]. For example, [10] extracts multisource data features using extended attribute profiles and then uses principal component analysis to reduce the stacked feature subspace dimension, followed by random forest (RF) or support vector machine for classification. Liao et al. [11] use a generalized graph-based fusion method for feature fusion and classification. In [12], extinction profiles (EPs) combined with orthogonal total variation component analysis are used to estimate high-dimensional features in a lower dimensional space to produce accurate classification maps. In [13], information is extracted from multiple sources of data separately using EPs and then fused to generate a classification map using sparse and low-rank subspace modeling. Jia et al. [14] used the RF to classify Gabor features from different data sources and then introduced a weighted majority voting-based decision fusion strategy to enhance the classification results. However, these traditional methods rely heavily on hand-designed features, which fail to adaptively capture the complete information in the data. Furthermore, the performance of the methods is influenced by the parameter settings, which need to be carefully tuned to achieve better results.

B. Deep-Learning-Based Methods

Compared with traditional methods, deep learning methods excel at automatically uncovering valuable information within the data without reliance on manually crafted features. Early attempts treated LiDAR as an additional spectral band of HSI, fed together into a CNN model for classification [16]. Xu et al. [17] constructed a two-branch CNN in which multisource features are combined in a cascading manner.

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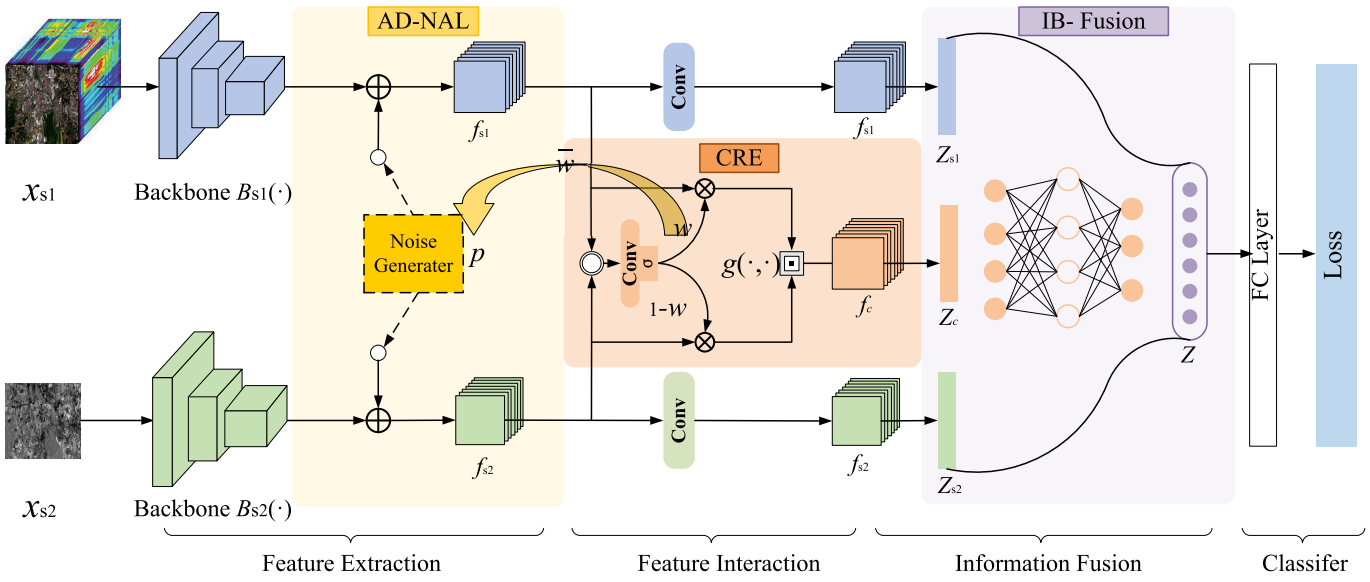


Fig. 1. Overview architecture of the MIBF-Net. The MIBF-Net is mainly composed of the following components. (i) Feature extraction. (ii) Feature interaction. (iii) IB-Fusion. (iv) Classifier.

In [18], both the feature-level and decision-level fusion methods are used to effectively combine the multisource features. Li et al. [19] proposed an innovative AsyFFNet that uses weight-share residual blocks and separates batch normalization layers for effective feature extraction. Gao et al. [20] introduced the adversarial complementary learning (ACL) strategy into the CNN model called ACL-CNN to extract the complementary information of the multisource data. These methods underperform in dealing with complex scenarios due to the lack of multisource information interaction. Liu et al. [22] introduced the deep multiview joint learning network (DMULN), including a CCA-based associated feature extractor and a view union pooling to extract and fuse multiview features for multisource data classification. However, the CCA-based correlation feature extractor does not take into account the vast feature differences in heterogeneous situations. Hong et al. [23] present an encoder–decoder-based architecture for combining multimodal features by reconstructing multimodal inputs. This method reconstructs the original data to obtain an information-rich representation, but the representation is inevitably redundant. Fang et al. [24] facilitated cross-modal information interaction between multisource data with an attention mechanism. In [26], a cross-scale mixing attention transformer is presented to tackle the challenge of extracting long-range dependencies. Xue et al. [27] developed a deep hierarchical vision transformer (DHViT) for multisource images' joint classification. But the above three methods do not consider the imbalance and inadequacy of heterogeneous feature extraction.

III. Method

In this article, we propose an MIBF-Net for multisource heterogeneous remote sensing image classification. We first introduce the overview of MIBF-Net, and then detail the design of three key modules: AD-NAL, CRE module, and IB-Fusion module.

A. Overview

The MIBF-Net enables the model to collaboratively learn source-specific information and cross-source interaction structures while reducing the impact of redundant information, ultimately obtaining a unified representation of multiple sources. The diagram of MIBF-Net is shown in Fig. 1, and the pipeline is illustrated in Algorithm 1. The design of the network architecture follows a dual-branch structure in which each branch corresponds to a data source. Specifically, there are four feature processing stages that perform feature extraction, interaction, fusion, and classification in sequence.

In the feature extraction stage, we use a combination of three convolutional and SELayer [38] structures to build the feature extraction backbone $B_{s1}(\cdot;\theta_{s1})$ and $B_{s2}(\cdot;\theta_{s2})$ parameterized by θ_{s1} and θ_{s2}, respectively. A pair of corresponding patches $x_{s1} \in \mathbb{R}^{c1\times r\times r}$ and $x_{s2} \in \mathbb{R}^{c2\times r\times r}$ from different source images are fed into the backbone to extracting different source features $f_{s1} = B_{s1}(x_{s1};\theta_{s1})$ and $f_{s2} = B_{s2}(x_{s2};\theta_{s2})$, respectively. Here, r represents the size of the image patches, and $c1$ and $c2$ represent the number of different source image channels, respectively. Following that, we apply AD-NAL to balance the feature extraction of the two branches (see Section III-B). In the interaction stage, the CRE module models the complex relationships between different heterogeneous features (see Section III-C). In the fusion phase, the IB-Fusion module fuses the previously obtained source-specific features and cross-source interactions into a unified representation z. This process is based on information theory, which effectively eliminates noise and redundancy (see Section III-D). In the classification stage, the multisource joint representation z is processed by multiple fully connected layers $D(\cdot;\theta_d)$ parameterized by θ_d to estimate the class probability $\hat{y} = D(z;\theta_d)$.

B. Attribution-Driven Noise Adaptive Layer

The feature extraction in a dual-branch multisource fusion classification framework may be suboptimal due to data

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Algorithm 1 MIBF-Net Framework

Input: Multisource remote sensing image dataset $X_{si} = \{x_{si}^1, x_{si}^2, \ldots, x_{si}^N\},\ i \in \{1, 2\}$ of training sample size N, training epoch E, hyperparameter β;

Output: Learned model parameters Θ

Initialize network Θ and $\bar{w} = 0.5$
for $epoch = 0, 1, \cdots, E-1$ **do**
  // Feature Extraction
  Extract features from the images via $f_{s1} = B_{s1}(x_{s1}; \theta_{s1}),\ f_{s2} = B_{s2}(x_{s2}; \theta_{s2})$
  Compute the balanced feature by (1) and (2) $f'_{s1}, f'_{s2} \leftarrow$**AD-NAL**$(f_{s1}, f_{s2}, \bar{w})$
  // Feature Interaction
  Compute the weight w in (3) and get $\bar{w}$
  Calculate cross-source feature by (4) $f_c \leftarrow$**CRE**(f'_{s1}, f'_{s2}, w)
  Unisource feature transform $f''_{s1} = Conv_3(f'_{s1}),\ f''_{s2} = Conv_3(f'_{s2})$
  // Information Fusion
  Flatten features $z_{s1}, z_{s2}, z_c = Flatten(f''_{s1}, f''_{s2}, f_c)$
  Calculate the fusion $f = F(z_{s1}, z_{s2}, z_c; \theta_f)$
  Encoder $\boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z = \boldsymbol{\mu}(f; \theta_\mu), \boldsymbol{\Sigma}(f; \theta_\Sigma)$
  Generate $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$
  Use the reparameterization trick get joint representation $z = \boldsymbol{\mu}_z + \boldsymbol{\Sigma}_z \times \boldsymbol{\epsilon}$
  // Classifer
  Decoder $\hat{y} = \text{Sigmoid}(D(z; \theta_d))$
  Minimize $L_{IB-fusion}$ as in (20) and update the network parameters Θ by the Adam optimizer
end
Return The network parameters Θ

heterogeneity and quality imbalances. For heterogeneous multisource fusion classification tasks such as HS and SAR, the HS image is of higher quality, while the SAR image tends to be noisy with lower quality and more challenging to represent. The joint discriminative loss typically prioritizes HS images because of their more easily extracted information. Meanwhile, SAR images, regarded as secondary data sources, may be neglected for feature optimization. This imbalance leads to inadequate extraction of SAR features and failure to fully exploit the benefits of multisource data.

Motivated by the above concerns, we introduce AD-NAL to mitigate the dependency on specific source data, thereby optimizing the feature extraction process' imbalance. Given $f_{s1} \in \mathbb{R}^{C\times H\times W}$ and $f_{s2} \in \mathbb{R}^{C\times H\times W}$ represent the convolutional features of different source data after source-specific feature encoder. The AD-NAL process can be formalized as follows:

$$f'_{s1} = \delta_t \cdot \varepsilon \oplus f_{s1} \tag{1}$$

$$f'_{s2} = (1 - \delta_t) \cdot \varepsilon \oplus f_{s2} \tag{2}$$

where $\oplus$ represents elementwise addition. $\varepsilon \in \mathbb{R}^{C\times H\times W}$ is a randomly generated noise from a normal distribution, i.e., $\varepsilon \sim \mathcal{N}(0, I)$. $\delta_t \in \{0, 1\}$ is a Bernoulli random variable with probability $q_t\ , q_t \geq 1$, i.e., $\delta_t \sim \text{Bernoulli}(q_t)$. The value of q_t determines the frequency of noise added to each data source, and it is obtained from the CRE module discussed in Section III-C.

In summary, AD-NAL enhanced the model's reliance on low attribution source features and facilitated the balanced extraction of features from different sources. Essentially, AD-NAL can be regarded as a regularization technique that introduces noise into the highly attributed source features and encourages the lower attributed source features to learn independent features.

C. Cross-Source Relationship Encoding Module

Most existing methods use linear strategies (e.g., concatenation, Hadamard product, and elementwise add) for fuse features from different data sources to obtain multisource joint information. We argue that these linear methods are insufficient for modeling complicated relationships between heterogeneous multisource data. This insufficiency hinders the effective integration of multisource information, ultimately impacting classification accuracy. To this end, we introduce a CRE module that models complex multisource feature relationships using weighted bilinear interactions.

To obtain better interaction information, a weighting factor $w \in \mathbb{R}^{C\times H\times W}$ is introduced, serving as an indicator of different source feature importance. Specifically, given the unisource features $f'_{s1} \in \mathbb{R}^{C\times H\times W}$ and $f'_{s2} \in \mathbb{R}^{C\times H\times W}$, we build a relative information weighting network to dynamically calculate weight factor w

$$w = \sigma\left(\text{Conv}_1 \downarrow_c \left(\left[f'_{s1}, f'_{s2}\right]\right)\right) \tag{3}$$

where $[\cdot, \cdot]$ denotes the channel dimension concatenation. $C_1 \downarrow_c$ denotes the channel dimension reduction using 1×1 convolution. $\sigma(\cdot)$ is the sigmoid function, which limits w to range $(0, 1)$. In the CRE module, the multisource features are weighted using the weight factor w. Through this approach, the model emphasizes task-relevant information, thereby enhancing the effectiveness of subsequent interactions. This is because the input features processed are more targeted and can precisely capture meaningful feature combinations for the task. The weighted single-source features are bilinearly interacted and then passed through a convolution layer to obtain feature f_c. This process can be specifically expressed as follows:

$$f_c = \text{Conv}_1 \uparrow_c \left(g\left(f'_{s1} \odot w, f'_{s2} \odot (1 - w)\right)\right) \tag{4}$$

where $\odot$ is the Hadamard product. $g(\cdot, \cdot)$ denotes the outer product. $C_1 \uparrow_c$ denotes a 1×1 convolution with increased channel dimension.

The outer product $g(\cdot, \cdot)$ is a simple way to implement bilinear interactions. It can be calculated for second-order feature interactions, and the value of each element in the resulting feature matrix can be interpreted as an interaction between features from different data sources. From the above equation, it can be seen that w is used as the weight of one of the data source features, while $(1 - w)$ is used as the weight of the other data source feature. The average value of w is

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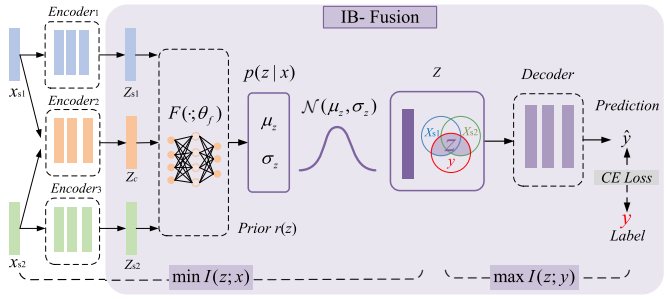


Fig. 2. Illustration of the IB-Fusion module.

calculated to obtain $\bar{w}$. The value of $\bar{w}$ reflects the relative importance of features from different data sources. We assign $\bar{w}$ to q_t in the AD-NAL module to guide which data source feature should be added to the noise.

It can be seen that the CRE module dynamically learns the feature importance and fine-grained feature interactions, which enhances the ability to capture feature interactions and nonlinear relationships. As a result, this approach enables a more effective exploration and understanding of higher order interactions and potential dependencies between heterogeneous multisource data.

D. Information Bottleneck Fusion Module

The information bottleneck (IB) principle aims to learn task-relevant representations by discarding as much task-irrelevant input information as possible, thereby promoting the robustness of the generated representations [39]. Since multisource representation is typically high-dimensional and contains redundant information [40], applying the IB principle to multisource representation is logical and helpful for prediction. We introduce the IB-Fusion module (as illustrated in Fig. 2), a new approach based on the IB principle, tailored to obtain a comprehensive and nonredundant multisource joint representation. The module integrates unisource information and cross-source interactions while filtering out redundant and noisy information, thereby enhancing the discriminative ability and noise tolerance of the joint representation.

1) Modeling: The IB-Fusion module aims to integrate knowledge from various data sources and learn a unified representation z. Given two multisource remote sensing images x_{s1} and x_{s2} with class label y. There are three key components that must be considered when modeling this joint representation z. The first component aims to maximize the mutual information between z and y, i.e., $\max I(z;y)$, ensuring that the learned joint representation is discriminative to the category labels and thus facilitating classification. The second component aims to minimize the mutual information between z and their original data x_{s1} and x_{s2}, denoted as $\min I(z;\{x_{s1},x_{s2}\})$. This objective effectively reduces redundancy and irrelevant information to ensure the learned joint representation remains concise. The third is a constraint term that uses a neural network to integrate single-source representations (z_{s1} and z_{s2}) and cross-source interaction representations (z_c) into a joint multisource representation z. Accordingly, the objectives are defined as

$$\begin{aligned} J_{\text{IB-Fusion}} &= \max_z I(z;y) - \beta I(z;\{x_{s1},x_{s2}\}) \\ &= \max_z I(z;y) - \beta I(z;x) \\ &\quad \text{s.t. } z = F(z_{s1},z_{s2},z_c;\theta_f) \end{aligned} \tag{5}$$

where β serves as a balancing coefficient which tunes the tradeoff between prediction and compression. The impact of varying β values will be explored in Section IV-C2. $F(\cdot;\theta_f)$ is the neural network parameterized by θ_f.

2) Optimization: The mutual information terms in the objective function defined in (5) are computationally intractable. To address this issue, we use the approach presented in variational information bottlenecks [41], which introduces a variational distribution that transforms the computationally intractable problem of computing mutual information into a more tractable one. Specifically, we infer a lower bound on the objective function by introducing a variance distribution, and this lower bound provides an approximation to the original objective function, as detailed in Theorem 1.

Theorem 1: Let $q(y|z)$ and $t(z)$ be the variational approximations to the true posterior distribution $p(y|z)$ and marginal distribution $p(z)$, respectively. Assuming $t(z)$ and $\boldsymbol{\epsilon}$ follow a standard normal Gaussian distribution. For a specific deep neural network, let N denote the batch size and $z = F(z_{s1},z_{s2},z_c;\theta_f)$. Assuming $p(z|x)$ is a Gaussian distribution learned by $\mathcal{N}(\boldsymbol{\mu}_z(x;\theta_\mu),\boldsymbol{\Sigma}_z(x;\theta_\Sigma)) = \mathcal{N}(\mu_z,\sigma_z)$. Under these conditions, the IB-Fusion objective $J_{\text{IB-Fusion}}$ is lower bounded by

$$\begin{aligned} &\frac{1}{N}\sum_{n=1}^{N}\Big[\mathbb{E}_{\boldsymbol{\epsilon}\sim p(\boldsymbol{\epsilon})}[\log q(y_n \mid z_n)] \\ &\qquad - \beta\cdot \mathrm{KL}\Big(\mathcal{N}(\mu_{z_n},\sigma_{z_n})||\mathcal{N}(0,I)\Big)\Big]. \end{aligned} \tag{6}$$

Proof of Theorem 1: For the first term $I(z;y)$ of $J_{\text{IB-Fusion}}$, according to the definition of mutual information, we have

$$\begin{aligned} I(z;y) &= \int dy\,dz\;\; p(y,z)\log\frac{p(y,z)}{p(y)p(z)} \\ &= \int dy\,dz\;\; p(y,z)\log\frac{p(y|z)}{p(y)}. \end{aligned} \tag{7}$$

Based on the nonnegativity property of the Kullback–Leibler (KL) divergence, we can derive the following results:

$$\begin{aligned} &\mathrm{KL}[p(y|z),q(y|z)] \geq 0 \\ &\Rightarrow \int dy\; p(y|z)\log p(y|z) \geq \int dy\; p(y|z)\log q(y|z) \end{aligned} \tag{8}$$

where $q(y|z)$ is a variational approximation to $p(y|z)$. And hence,

$$I(z;y) \geq \int dy\,dz\;\; p(y,z)\log\frac{q(y|z)}{p(y)} \tag{9}$$

$$= \int dy\,dz\;\; p(y,z)\log q(y|z) + H(y) \tag{10}$$

$$\geq \int dx\,dy\;\; dz\;\; p(z|x)p(y|x)p(x)\log q(y|z). \tag{11}$$

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The term $H(y)$ represents the entropy of the label, which can be treated as a constant and ignored during parameter optimization. Consequently, we can focus on maximizing the lower bound, as shown in (11), to optimize the objective function $I(z; y)$.

We now consider the second term $I(z; x)$ of $J_{\text{IB-Fusion}}$, which represents the minimal information constraint. Taking into account that the KL divergence is a nonnegative quantity, we can obtain

$$\begin{aligned} I(z;x) &= \int dz\, dx \;\; p(x,z) \log \frac{p(z|x)}{p(z)} \\ &\leq \int dx\, dz \;\; p(x)p(z|x) \log \frac{p(z|x)}{t(z)} \end{aligned} \tag{12}$$

where $t(z)$ is a variational approximation of $p(z)$.

Combining the above two bounds, we have that

$$\begin{aligned} &J_{\text{IB-Fusion}} \\ &\geq F_{\text{IB-Fusion}} \\ &= \mathbb{E}_{(x,y)\sim p(x,y), z\sim p(z|x)} \Big[\log q(y|z) - \beta \cdot \text{KL}\Big(p(z|x) || t(z) \Big) \Big]. \end{aligned} \tag{13}$$

Here, $F_{\text{IB-Fusion}}$ represents a lower bound of $J_{\text{IB-Fusion}}$. By maximizing $F_{\text{IB-Fusion}}$, we can effectively facilitate the optimization of $J_{\text{IB-Fusion}}$.

To optimize $F_{\text{IB-Fusion}}$ with deep neural networks, we assume that $p(z|x)$ is a Gaussian distribution, the mean and variance of which can be learned through deep neural networks

$$p(z \mid x) = \mathcal{N}\big(\boldsymbol{\mu}_z(x;\theta_\mu), \boldsymbol{\Sigma}_z(x;\theta_\Sigma)\big) = \mathcal{N}\big(\mu_z, \sigma_z\big) \tag{14}$$

where the deep neural networks $\boldsymbol{\mu}_z(\cdot;\theta_\mu)$ and $\boldsymbol{\Sigma}_z(\cdot;\theta_\Sigma)$ are parameterized by θ_μ and θ_Σ, respectively, to learn the mean μ_z and variance σ_z of Gaussian distributions. We can use the reparameterization trick to obtain z

$$z = \mu_z + \sigma_z \times \boldsymbol{\epsilon} \tag{15}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$.

Furthermore, assume $t(z)$ follows a standard normal Gaussian distribution, denoted by $\mathcal{N}(0, I)$. Combining (14), the second term of (13) can be formulated as

$$\text{KL}\Big(p(z \mid x) || t(z) \Big) = \text{KL}\Big(\mathcal{N}\big(\mu_z, \sigma_z\big) || \mathcal{N}(0, I) \Big). \tag{16}$$

Finally, we can simplify $F_{\text{IB-Fusion}}$ by approximating the integral over x, z, and y using Monte Carlo sampling [42]

$$\begin{aligned} &F_{\text{IB-fusion}} \\ &\approx \frac{1}{N} \sum_{n=1}^{N} \Big[\mathbb{E}_{\boldsymbol{\epsilon}\sim p(\boldsymbol{\epsilon})} [\log q(y_n \mid z_n)] \\ &\qquad - \beta \cdot \text{KL}\Big(p(z_n \mid x_n) || q(z_n) \Big) \Big] \\ &= \frac{1}{N} \sum_{n=1}^{N} \Big[\mathbb{E}_{\boldsymbol{\epsilon}\sim p(\boldsymbol{\epsilon})} [\log q(y_n \mid z_n)] \\ &\qquad - \beta \cdot \text{KL}\Big(\mathcal{N}(\mu_{z_n}, \sigma_{z_n}) || \mathcal{N}(0, I) \Big) \Big] \end{aligned} \tag{17}$$

where N represents the batch size. ■

According to Theorem 1, the IB-Fusion module's procedure can be clearly obtained. Specifically, the representations z_{s1}, z_{s2}, and z_c are encoded as intermediate representations f. Then, according to (14), take f as an input to the fully connected layer to generate μ_z and σ_z. We use the reparameterization technique to obtain the joint representation z by (15).

The first term in the objective function (6) corresponds to the cross-entropy loss. To calculate the category probability $\hat{y}$ and the loss, we can use a fully connected layer with a Softmax layer

$$\hat{y} = \text{Softmax}(D(z;\theta_d)) \tag{18}$$

$$L_{\text{CE}} = -\frac{1}{N} \sum_{n=1}^{N} \left(y_n \log(\widehat{y}_n) + (1 - y_n) \log(1 - \widehat{y}_n) \right) \tag{19}$$

where $D(\cdot;\theta_d)$ is a fully connected layer parameterized by θ_d.

The second term in the objective function (6) is the KL divergence, which is calculated analytically to obtain the exact result [43]. The resulting total loss for multisource fusion classification can be expressed as

$$L_{\text{IB-Fusion}} = L_{\text{CE}} - \beta \frac{1}{2} \sum_{i=1}^{M} (1 + \log((\sigma_{z_i})^2) - (\sigma_{z_i})^2 - (\mu_{z_i})^2) \tag{20}$$

where M denotes the dimensionality of the z, while μ_i and σ_i represent the ith elements used to generate z.

By optimizing this objective function, the irrelevant information in the original multisource data can be reduced, and we can obtain a nonredundant and comprehensive multisource joint representation.

IV. Experiments

To evaluate our proposed method, we perform experiments on three prevalent heterogeneous multisource remote sensing datasets. We first present the dataset details, followed by the experimental setup description and ablation studies. Finally, we compare our approach with the state-of-the-art deep-learning-based methods to demonstrate its superiority.

A. Dataset Description

1) MUUFL Dataset: The MUUFL dataset [44] was captured in November 2010 at the Gulfport Campus of the University of Southern Mississippi, Long Beach, MS, USA. The data collection includes hyperspectral images (HSIs) and a digital surface model based on LiDAR. The HSI is captured using the ITRES Compact Airborne Spectral Imager sensor and consists of 64 spectral channels ranging from 0.38 to 1.05 μm. LiDAR data are obtained using the ALTM sensor and comprise two bands. The dataset's spatial dimensions are 325 × 220, and it features a 1-m spatial resolution. Ground truth for this dataset covers 11 categories, with 53 687 labeled samples. During the training process, we randomly select 2% of the data in the dataset for training and the rest for testing.

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2) Houston 2013 Dataset: The dataset was collected using the Compact Airborne Spectral Imager sensor at the University of Houston, Houston, TX, USA, and its surrounding areas in 2012. It was released in the 2013 IEEE GRSS (IGRSS) contest [45] and consists of two sources of data: HSI and LiDAR-derived digital surface models. The hyperspectral imagery covers a spectral wavelength range of 364–1046 nm and comprises 144 spectral bands, while the LiDAR data consist of only one band. Both data types have a spatial resolution of 2.5 m and spatial dimensions of 349 × 1905. The dataset includes 15 categories. For our experiment, we create a training set by randomly selecting 20 points from each category and using the remaining points to form the test set.

3) Houston 2018 Dataset: The Houston 2018 dataset, covering the University of Houston campus, was publicly released in the 2018 IGRSS contest [46]. This dataset comprises LiDAR data, HSI, and very high-resolution RGB imagery remote sensing data. The HSI consists of 48 spectral bands, with a spectral range spanning from 380 to 1050 nm. We resample the HSI, LiDAR data, and the ground truth at a 0.5-m GSD spatial resolution to standardize the spatial size of each image to 1202 × 4768 pixels. During network training, we randomly select 2% of the data from the dataset for training while the remaining data are allocated for testing.

B. Implementation Details

1) Experimental Settings: In this article, the implementation setup is based on the PyTorch Toolkit, and the core program is written in Python language. All the experiments are conducted on an HP-Z820 workstation configured with a GeForce GTX 1080 8-GB GPU under Ubuntu 18.04 LTS. There are no pretraining parameters, and all the parameters are initialized through Xavier [47]. The Adam optimizer [48] is set with a weight decay of 5^{-4}. And the batch_size is set to 64 by default. The initial learning rates for the MUUFL, Houston 2013, and Houston 2018 datasets are 0.001, 0.0005, and 0.002, respectively. The experiments use a cosine decay learning rate schedule with warmup epochs 10. We use data augmentation techniques such as random noise addition, flipping, and rotating training images. To evaluate the classification performance more fairly, the same training and test samples are applied to all the algorithms on each benchmark dataset. All the experiments are repeated five times to eliminate bias in deep learning model training, and the quantitative results are obtained by taking the average.

2) Evaluation Metrics: In the experiments, we evaluate the classification performance of various methods using both quantitative and qualitative approaches. Commonly used evaluation metrics such as overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are used to quantitatively evaluate the consistency of the predicted results with the ground truth. Higher values for these metrics indicate better classification performance. We also evaluate the classification performance from a visual perspective by creating classification maps through visualization. These maps allow for a qualitative evaluation of the classification performance.

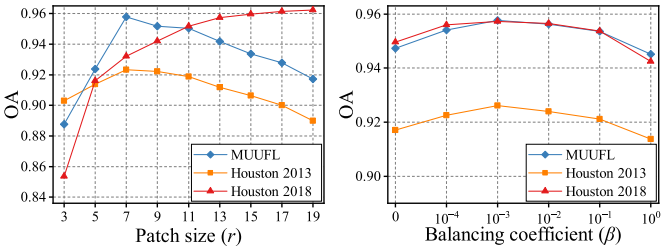


Fig. 3. (a) Effects of the image patch size on classification performance. (b) Effects of the balancing coefficient on classification performance.

C. Parameter Tuning and Module Analysis

1) Analysis of the Image Patch Size: The impact of different input patch sizes r on the model's performance is compared by evaluating the OA. Fig. 3(a) shows the classification performance of three datasets with input patch sizes r ranging from 3 × 3 to 19 × 19. What can be observed for the MUUFL and Houston 2013 datasets is that as the patch size increases, the classification performance initially improves and then declines. The patch size of 7 × 7 yields the optimal results on the two datasets. This is because smaller patches may lack sufficient contextual information to achieve accurate discrimination, while larger patches can have redundant information that negatively impacts the discriminability of the central pixel, resulting in reduced experimental performance. For the Houston 2018 dataset, classification performance increases as patch size increases, owing to the relatively higher complexity of the scenario. However, too large image patch sizes should be avoided, which can introduce excessive redundancy and affect the algorithm's efficiency. Specifically, we select 13 × 13 as the input size for the Houston 2018 dataset because the increase in classification accuracy tends to level off when the patch size of the training images exceeds 13. The results illustrate the necessity of choosing the appropriate input patch size for the proposed method.

2) Analysis on the Balancing Coefficient: The balancing coefficient β is a positive parameter that controls the tradeoff between the compression of potential representations and the preservation of categorical information [see (5)]. The effect of β on the three datasets is depicted in Fig. 3(b), and all of them display a similar trend: excessively small or large values of β result in a reduction in accuracy. Therefore, a moderate β value is essential to achieving optimal classification performance. It can be found that the proposed MIBF-Net achieves the best performance on the three datasets when $\beta = 10^{-3}$. Assigning excessively large values to β implies giving more weight to information compression, which will weaken the discriminative of the joint representation and lead to a decrease in classification performance. It is worth noting that setting β to zero results in a decrease in accuracy of at least 0.77% compared with the performance achieved with $\beta = 10^{-3}$. In fact, with β being zero, the minimum information constraint

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TABLE I
ABLATION ANALYSES OF DIFFERENT COMPONENTS ON THREE DATASETS

Method	Modules						MUUFL			Houston 2013			Houston 2018		
	AD-NAL	Random-NAL	CRE	Sum	Multiply	IB-Fusion	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Proposed(w/o AD-NAL)			✓			✓	91.61	79.22	88.82	94.53	95.12	94.09	95.07	93.02	93.18
Proposed(w/o AD-NAL*)		✓	✓			✓	92.05	79.72	89.56	95.22	95.80	94.83	95.32	93.18	93.66
Proposed(w/o CRE)	✓					✓	91.25	78.32	88.53	93.46	94.55	92.93	94.63	92.65	93.82
Proposed(w/o CRE*)	✓			✓		✓	91.72	79.88	89.31	95.14	95.82	94.75	95.12	93.11	94.08
Proposed(w/o CRE†)	✓				✓	✓	91.68	79.46	88.98	94.98	95.73	94.58	94.83	92.92	93.88
Proposed(w/o IB-Fusion)	✓		✓				91.49	79.03	89.17	94.73	95.35	94.89	94.96	93.38	93.52
Proposed	✓		✓			✓	**92.61**	**81.96**	**90.24**	**95.77**	**96.23**	**95.43**	**95.73**	**93.41**	**94.45**

* Proposed(w/o ⋆) represent the MIBF-Net without module ⋆.

TABLE II
CLASSIFICATION ACCURACY (%) ON THE MUUFL DATASET WITH 2% SAMPLES

NO.	Class(Train/Test)	TB_CNN	Endnet	MDL_RS	FusAtNet	S2ENet	MFT	DFINet	CALC	MIBF-Net
1	Tree (465/22781)	96.46±0.59	92.86±1.75	94.64±1.23	96.99±0.44	95.76±0.79	96.41±0.63	96.44±0.60	96.10±0.33	**97.01±0.45**
2	Mostly grass (86/4184)	77.82±3.34	72.58±3.54	82.35±2.61	77.20±3.51	74.86±2.76	75.55±2.49	78.43±2.85	78.62±2.44	**79.44±2.28**
3	Mixed ground (138/6744)	79.84±2.95	71.82±4.25	69.61±5.83	83.85±1.85	82.78±0.92	79.82±1.90	81.04±1.21	80.91±1.71	**84.86±1.36**
4	Dirt and sand (37/1789)	90.98±0.84	76.47±3.88	93.45±1.29	83.96±1.24	93.87±0.05	90.43±1.26	89.82±0.87	85.08±1.08	**93.98±0.82**
5	Road (134/6553)	93.73±0.95	91.12±1.24	92.58±1.75	94.33±0.83	93.51±0.16	91.33±1.07	94.01±0.86	91.26±0.94	**95.15±0.61**
6	Water (10/456)	86.31±1.56	**97.94±0.62**	97.70±0.56	97.23±0.50	96.93±0.76	95.58±0.89	92.93±1.64	93.47±1.25	95.80±0.87
7	Building (45/2188)	84.24±1.40	78.06±3.09	**88.59±1.05**	83.78±1.86	83.72±1.74	82.42±1.92	88.54±1.80	86.85±1.81	84.31±1.32
8	Buliding (125/6115)	96.58±0.76	96.18±1.31	97.09±0.63	97.66±0.47	97.52±0.88	97.39±0.60	96.52±0.88	96.14±1.63	**98.23±0.34**
9	Sidewalk (28/1357)	73.79±3.77	58.17±3.87	**82.37±3.45**	69.84±4.58	78.07±3.22	58.86±3.26	67.50±3.91	63.47±3.15	79.19±2.07
10	Yellow curb (4/179)	33.63±2.68	28.88±3.11	12.15±5.59	23.07±3.18	16.48±4.31	15.62±4.25	23.33±3.85	18.42±3.65	**38.80±2.58**
11	Cloth panels (6/263)	76.86±3.45	80.02±1.39	78.98±1.86	74.07±1.95	76.30±1.60	77.53±1.97	81.24±1.34	78.08±1.80	**81.56±1.16**
	OA(%)	90.11±1.78	86.68±2.32	89.53±2.16	90.81±1.14	91.30±0.93	90.42±1.02	91.04±1.00	91.23±1.31	**92.61±0.75**
	AA(%)	74.81±2.20	68.65±2.65	68.18±2.48	76.92±1.92	76.54±1.64	73.63±1.83	76.27±1.72	75.91±1.58	**80.96±1.30**
	Kappa(%)	88.23±1.94	82.25±2.38	86.02±2.20	89.15±1.38	89.19±1.07	87.32±1.13	89.22±1.10	88.50±1.35	**90.24±0.87**

is removed, and the framework degenerates into a classification framework. The decline in performance indicates the effectiveness of the introduced IB-Fusion module in improving the performance.

3) Impact of Attribution-Driven Noise Adaptive Layer: The performance of AD-NAL was further studied through ablation experiments. As shown in Table I, when MIBF-Net removes the AD-NAL, classification accuracy is significantly degraded. This is due to the fact that unbalanced information extraction degrades the classification performance. Taking the MUUFL dataset as an example, the OA of the proposed (w/o AD-NAL) is only about 91.61%. When the random noise adaptive layer is introduced, the classification accuracy of the proposed (w/o AD-NAL*) can be improved by about 0.44%, which indicates that the random noise alleviates the optimization imbalance problem to some extent. For the proposed method with the AD-NAL module, the classification accuracy is further improved by 0.56%. This ablation fully illustrates the effectiveness of the AD-NAL.

4) Impact of Cross-Source Relationship Encoding Module: Further ablation studies were conducted to demonstrate the effectiveness of the proposed CRE module. Specifically, the results are given in Table I. The proposed (w/o CRE) method means removing the feature interaction module and transferring only the individual features of the data source. There is a significant performance degradation (at least 1.1%) on all three datasets due to the inability to capture the complex relationships between multiple data sources. We also use two different feature interaction methods in the experiment for a more careful comparison. The proposed (w/o CRE*) refers to the elementwise addition. The proposed (w/o CRE†) method refers to the Hadamard product. It can be seen from the classification results that the gated bilinear feature interaction has higher classification accuracy. According to the experimental results, the feature interaction method based on gated bilinear features can fully mine the interaction patterns between heterogeneous features and effectively fuse the information of different sources. This illustrates the importance of higher order interaction of information across data sources.

D. Results and Comparisons

We compare MIBF-Net with seven state-of-the-art methods in the experiments to verify the effectiveness of the proposed method, including TB_CNN [17], Endnet [23], MDL_RS [15], FusAtNet [49], S2ENet [24], MFT [50], DFINet [51], and CALC [52]. All the comparative methods use default parameters from their original papers to ensure the best results' presentation.

Tables II–IV report the detailed experimental results of the different methods in terms of OA, AA, Kappa, and per-class

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TABLE III
CLASSIFICATION ACCURACY (%) ON THE HOUSTON 2013 DATASET WITH 20 SAMPLES PER CLASS

NO.	Class(Train/Test)	TB_CNN	Endnet	MDL_RS	FusAtNet	S2ENet	MFT	DFINet	CALC	MIBT-Net
1	Health grass (20/1231)	90.28±1.77	95.11±1.07	98.45±0.67	94.28±0.80	97.08±0.43	**97.45±0.51**	97.14±0.60	90.10±0.91	96.32±0.50
2	Stressed grass (20/1234)	92.67±1.31	90.73±1.29	92.34±1.55	94.82±0.49	99.25±0.19	91.73±0.83	98.98±0.48	96.08±0.27	**99.42±0.37**
3	Synthetic grass (20/677)	90.85±1.26	83.75±1.14	94.60±0.79	91.82±1.06	94.79±0.92	96.53±0.67	94.61±0.73	**96.63±0.39**	93.66±0.88
4	Trees (20/1224)	93.87±0.89	95.47±0.87	96.75±0.81	97.18±0.71	97.82±0.74	93.47±0.91	96.98±0.54	95.58±0.88	**98.05±0.71**
5	Soil (20/1222)	97.62±0.32	94.51±0.90	97.44±0.60	97.99±0.62	93.71±0.83	96.75±0.45	96.44±0.29	**98.88±0.68**	97.68±0.92
6	Water (20/305)	92.28±1.56	96.35±0.42	98.32±0.48	96.20±0.73	**100.0±0.00**	92.28±0.85	91.18±0.94	94.17±0.19	99.32±0.25
7	Residential (20/1248)	95.69±1.19	86.14±1.11	**96.32±0.56**	87.12±1.23	95.31±0.77	90.62±0.23	94.33±0.63	95.12±0.14	95.59±0.66
8	Commercial (20/1224)	88.92±1.42	83.48±2.30	82.89±1.24	89.77±2.01	97.99±0.60	98.01±0.46	98.10±0.60	92.78±0.82	**98.19±0.79**
9	Road (20/1232)	91.50±1.31	72.34±1.49	74.47±1.84	88.58±1.66	86.05±2.37	91.67±1.37	92.36±0.42	89.77±0.27	**92.86±0.58**
10	Highway (20/1207)	90.74±1.19	66.60±2.90	**93.99±0.63**	92.57±0.99	92.88±1.33	89.52±0.89	91.01±0.81	93.03±0.59	91.35±1.02
11	Railway (20/1215)	93.45±0.70	71.33±1.87	89.09±1.75	**93.95±0.52**	88.56±1.38	93.90±0.77	92.07±0.99	93.82±0.61	92.50±0.45
12	Parking lot 1 (20/1213)	80.97±1.98	61.93±2.85	90.13±1.49	91.33±1.10	94.49±0.73	87.15±1.49	93.92±0.40	90.63±0.94	**94.95±0.25**
13	Parking lot 2 (20/449)	91.73±1.63	49.47±3.16	**96.80±0.93**	76.35±2.06	91.35±0.98	83.73±1.68	95.58±0.48	95.27±0.55	95.77±0.67
14	Tennis court (20/408)	96.91±0.89	84.84±1.89	99.27±0.34	93.57±1.18	**100.0±0.00**	97.60±0.61	96.91±0.66	99.90±0.04	97.37±0.36
15	Running track (20/640)	98.91±0.44	98.60±1.02	99.53±0.20	**99.84±0.11**	98.91±0.82	99.37±0.35	97.63±0.37	98.56±0.29	97.70±0.59
	OA(%)	91.49±1.42	82.25±2.26	92.01±0.95	93.17±0.87	94.05±0.79	93.31±0.98	94.41±0.77	94.70±0.70	**95.77±0.75**
	AA(%)	92.10±1.18	82.96±1.57	93.14±0.81	93.52±0.68	95.48±0.51	94.51±0.83	95.35±0.64	95.12±0.63	**96.23±0.56**
	Kappa(%)	90.87±1.22	80.82±1.72	91.36±0.85	92.62±0.72	93.81±0.54	92.77±0.84	93.95±0.71	94.26±0.65	**95.43±0.60**

TABLE IV
CLASSIFICATION ACCURACY (%) ON THE HOUSTON 2018 DATASET WITH 2% SAMPLES

NO.	Class(Train/Test)	TB_CNN	Endnet	MDL_RS	FusAtNet	S2ENet	MFT	DFINet	CALC	MIBF-Net
1	Healthy grass (784/38412)	**87.33±1.72**	85.08±1.91	87.55±1.60	75.39±2.25	79.46±2.46	87.26±1.11	80.09±2.12	85.98±1.88	85.67±1.85
2	Stressed grass (2601/127407)	93.19±1.30	89.83±1.76	95.07±0.91	96.39±0.94	**96.89±0.84**	96.00±0.87	96.84±0.83	95.77±0.91	96.39±0.66
3	Artificial turf (55/2681)	99.81±0.15	99.32±0.37	100.0±0.00	98.27±0.26	99.44±0.29	100.0±0.00	100.0±0.00	100.0±0.00	**100.0±0.00**
4	Evergreen trees (1087/53235)	94.01±1.52	93.41±0.87	96.35±1.21	95.23±1.14	95.28±0.90	95.65±0.63	95.34±0.61	94.71±0.66	**97.30±0.83**
5	Deciduous trees (404/19768)	83.78±1.74	82.12±1.92	90.63±1.73	87.47±1.58	90.15±1.88	91.08±1.92	90.00±1.52	86.73±1.23	**92.02±0.77**
6	Bare earth (362/17702)	94.79±1.19	81.61±1.74	98.38±0.52	97.09±0.42	97.04±0.68	**98.68±0.43**	97.32±0.49	98.39±0.45	98.65±0.29
7	Water (22/1042)	99.14±0.73	99.89±0.07	100.0±0.00	92.88±1.19	100.0±0.00	99.09±0.28	99.90±0.05	99.73±0.22	**100.0±0.00**
8	Residential buildings (3180/155815)	91.27±1.59	80.95±1.99	96.17±0.62	95.64±0.73	94.16±0.82	95.34±0.73	94.52±0.88	**97.56±0.40**	96.68±1.01
9	Non-residential buildings (17896/876873)	98.10±0.92	94.67±1.75	95.02±0.79	98.68±0.81	98.63±0.54	98.76±0.83	98.82±0.60	97.47±0.54	**99.22±0.36**
10	Roads (3666/179617)	78.34±1.98	62.53±3.42	84.05±1.22	81.66±2.67	84.92±1.28	88.08±1.52	84.79±1.76	87.25±0.82	**89.99±0.77**
11	Sidewalks (2721/133314)	74.09±2.35	65.52±2.87	83.63±1.54	82.31±2.39	85.20±2.41	83.57±1.97	83.73±1.47	82.99±1.61	**89.17±1.28**
12	Crosswalks (122/5937)	44.36±3.10	50.05±3.18	41.28±3.62	44.39±3.50	**52.24±2.70**	52.13±2.51	52.10±2.11	45.72±2.79	48.56±2.79
13	Major thoroughfares (3709/181729)	76.08±2.37	56.48±2.63	78.18±2.30	83.12±1.96	83.06±1.91	83.87±2.44	84.22±1.36	87.23±0.82	**89.30±1.41**
14	Highways (789/38649)	87.60±1.27	79.73±2.29	89.64±1.31	91.96±1.57	92.00±1.78	88.97±1.87	92.39±0.51	93.57±0.77	**96.69±0.53**
15	Railways (555/27193)	97.49±0.61	93.81±0.97	98.04±0.65	97.69±0.34	96.17±0.56	**98.70±0.39**	96.94±0.29	93.69±0.31	97.96±0.74
16	Paved parking lots (919/45013)	92.02±0.89	87.76±1.15	96.28±0.78	94.26±0.73	95.55±0.51	96.63±0.79	94.52±0.87	**96.80±0.78**	96.02±0.42
17	Unpaved parking lots (12/575)	10.02±4.78	**95.13±0.57**	12.05±5.25	25.21±4.11	33.60±2.61	31.54±3.22	29.65±3.26	32.27±2.92	92.12±0.96
18	Cars (526/25763)	93.79±1.41	76.72±2.60	94.81±0.96	94.17±0.97	95.05±0.82	95.06±0.85	94.91±1.34	96.20±0.45	**97.02±0.52**
19	Trains (430/21049)	98.04±0.87	92.57±1.23	98.51±0.74	98.74±0.25	99.25±0.60	98.99±0.27	98.87±0.31	99.39±0.38	**99.41±0.37**
20	Stadium seats (546/26750)	99.14±0.56	98.42±0.79	**100.0±0.00**	98.70±0.58	99.52±0.28	99.79±0.20	99.84±0.09	98.61±0.84	99.81±0.12
	OA(%)	90.85±1.38	83.37±1.41	91.76±1.02	93.05±1.31	92.71±0.99	93.55±0.89	93.37±0.77	93.12±0.62	**95.73±0.71**
	AA(%)	82.16±1.89	72.58±1.85	81.90±1.38	85.77±1.57	85.51±1.21	87.12±1.25	86.53±1.03	86.34±1.07	**93.11±0.89**
	Kappa(%)	88.11±1.42	78.27±1.50	89.17±1.06	90.97±1.34	90.82±1.02	92.44±0.92	91.39±0.80	91.02±0.83	**94.45±0.73**

precision. Where rows arrange the class precision and global metrics, and the methods considered are listed in columns. The bold values in these tables indicate the optimal values in the corresponding rows. In addition to the quantitative evaluation using evaluation metrics, we also use the classification graphs obtained from the different models for visual comparison. Figs. 4–6 show the classification maps obtained by the different methods on the three benchmark datasets, respectively. The classification maps displayed in these figures use different colors to represent different classes. In addition, the ground-truth maps are included for better comparison and evaluation of the classification maps. A specific analysis of the results for each dataset is presented below.

1) Results on MUUFL Dataset: Table II illustrates the experimental quantitative results on the MUUFL dataset. It can be concluded from the table that the proposed method outperforms other baseline methods in all the three indicators. To be specific, the proposed method has the best OA (92.61%) and reached the highest precision in eight classes. Meanwhile, compared with the second-best algorithm S2ENet, AA, OA,

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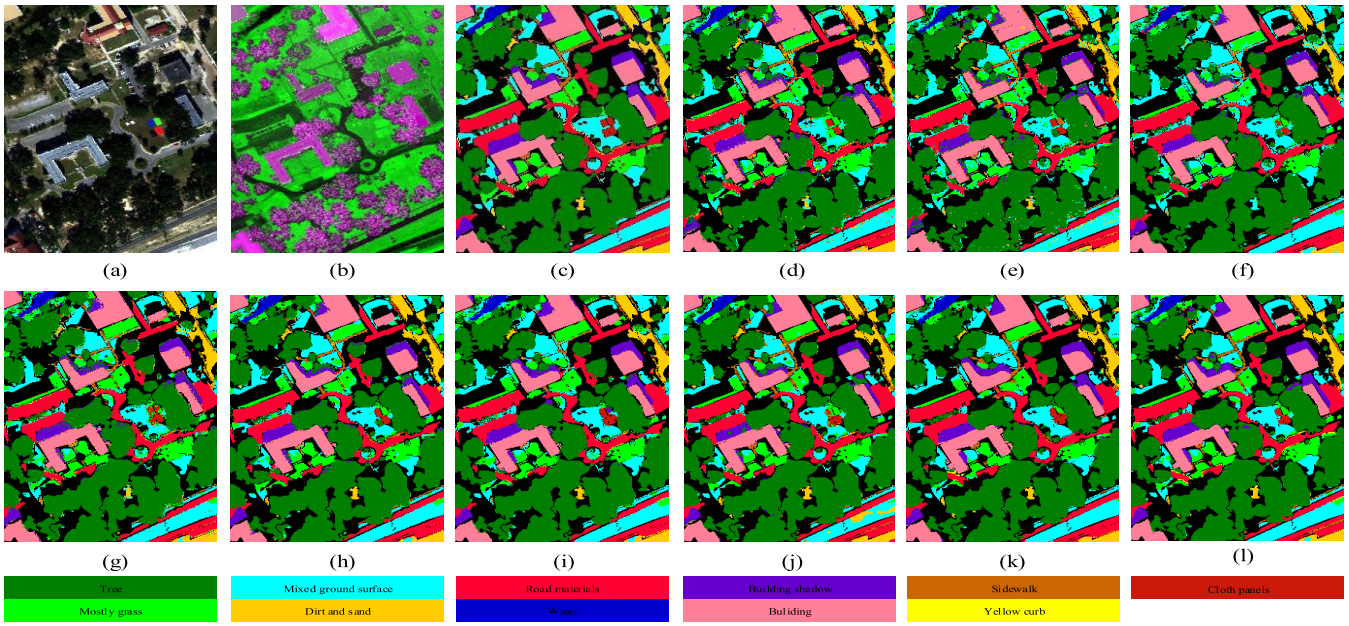


Fig. 4. Classification maps of different methods on the MUUFL dataset. (a) Hyperspectral pseudocolor image. (b) Visualization of LiDAR data. (c) Ground-truth map. (d) TB_CNN. (e) Endnet. (f) MDL_RS. (g) FusAtNet. (h) S2ENet. (i) MFT. (j) DFINet. (k) CALC. (l) Proposed.

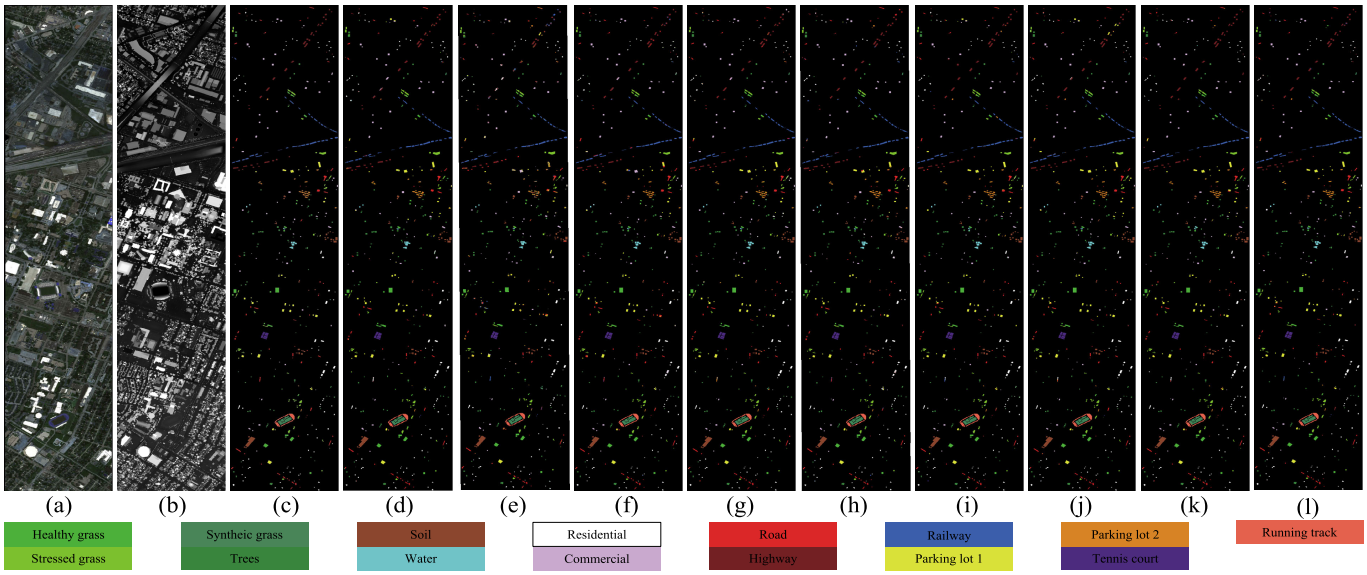


Fig. 5. Classification maps of different methods on the Houston 2013 dataset. (a) Hyperspectral pseudocolor image. (b) Visualization of LiDAR data. (c) Ground-truth map. (d) TB_CNN. (e) Endnet. (f) MDL_RS. (g) FusAtNet. (h) S2ENet. (i) MFT. (j) DFINet. (k) CALC. (l) Proposed.

and Kappa increased by 1.31%, 4.42%, and 1.05%, respectively. The visualized results of different methods are shown in Fig. 4. The classification maps demonstrate that the proposed method produces the most accurate and clean results, with clearly distinguished land cover categories and minimal noise.

2) Results on Houston 2013 Dataset: The Houston 2013 dataset's quantitative analysis results for each classification method are presented in Table III. The proposed method maintains the highest classification accuracy across all the three evaluation metrics. Concretely, AA is 95.77%, OA is 96.23%, and Kappa is 95.43%. The OA of MIBF-Net is better than the attention-based models S2ENet, MFT, and DFINet (i.e., 94.05%, 93.31%, and 94.41%), respectively, improved by 1.72%, 2.46%, and 1.36%, respectively. Furthermore, the classification performance exhibits excellent stability, as all the classes achieve precision exceeding 90%. Fig. 5 displays the classification maps generated by various methods. We can observe that the proposed method yields a very smooth resulting map that exhibits fewer mislabeled areas. It demonstrates better material discrimination capabilities than other methods, for example, trees, commercial, and road.

3) Results on Houston 2018 Dataset: The quantitative analysis results of each classification method on the Houston 2018 dataset are given in Table IV. The proposed method achieved the highest accuracies (i.e., OA = 95.73%, AA = 93.41%, and Kappa = 94.45%), obtaining at least 2.18%,

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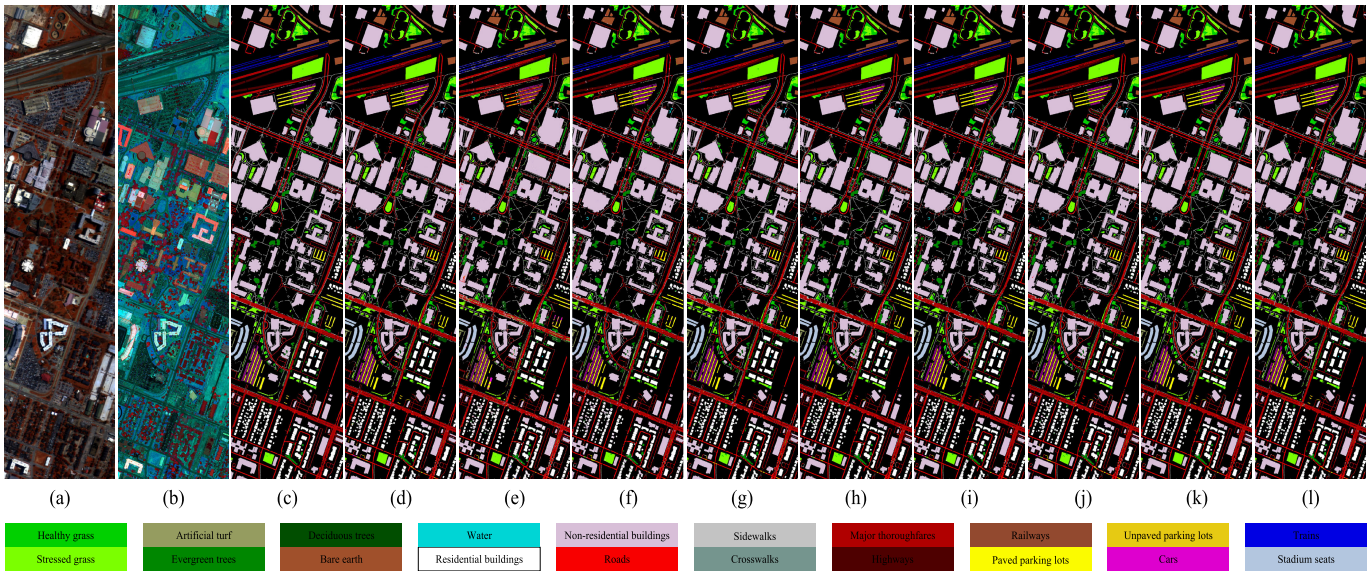


Fig. 6. Classification maps of different methods on the Houston 2018 dataset. (a) Hyperspectral pseudocolor image. (b) Visualization of LiDAR data. (c) Ground-truth map. (d) TB_CNN. (e) Endnet. (f) MDL_RS. (g) FusAtNet. (h) S2ENet. (i) MFT. (j) DFINet. (k) CALC. (l) Proposed.

TABLE V
COMPARISON ON THE NUMBER OF PARAMETERS AND FLOPS

Dataset	Complexity	TB_CNN	Endnet	MDL_RS	FusAtNet	S2ENet	MFT	DFINet	CALC	Proposed
MUUFL	Params(M)	3.864	0.089	0.107	36.26	0.178	0.249	0.898	0.834	0.284
	FLOPs(M)	128.8	0.173	3.56	6770	17.49	22.09	82.68	60.72	15.94
Houston2013	Params(M)	4.714	0.091	0.119	36.90	0.271	0.313	1.082	0.903	0.377
	FLOPs(M)	159.9	0.179	4.67	6930	26.51	50.61	112.3	95.67	24.96
Houston2018	Params(M)	3.697	0.089	0.106	36.15	0.160	0.238	0.864	0.825	0.269
	FLOPs(M)	122.8	0.174	3.36	6740	38.76	16.64	77.08	54.53	14.17

5.99%, and 2.01% improvement compared with the other methods, respectively. Especially for "Sidewalks" and "Highways," our method achieves 89.17% and 96.69%, which is far ahead of other methods. As can be seen from the visualization in Fig. 6, the proposed method shows smoother results for car classification and can clearly distinguish between Sidewalks and Highways, demonstrating its strong capability for fine-grained feature representation.

E. Model Complexity Analysis

Model complexity can be evaluated by different indicators, such as time complexity and space complexity, which measure the model's complexity in terms of time and space, respectively. In this work, we analyze the time complexity using the number of floating-point operations (FLOPs) and the spatial complexity using the number of model parameters. Table V presents the detailed FLOPs and model parameters of the different models. From the table, we can observe that the Endnet method can train the least parameters, while the FusAtNet uses the most parameters. This is due to the fact that Endnet has no need for high-dimensional feature mapping and learning, but the consequence is unsatisfactory classification performance. TB_CNN is designed with a deep and wide cascade-block structure, which improves the performance of the model while increasing the computational cost and resulting in a significant time overhead. When taking into account both computational cost and performance evaluation metrics, the MIBF-Net remains the competitive option among multisource data classification methods.

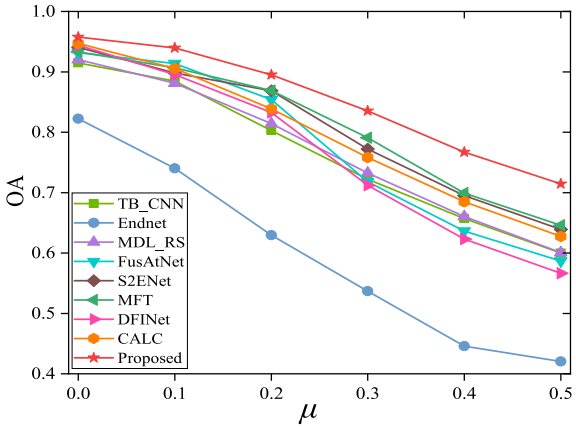


Fig. 7. Comparison of the robustness performance of various models on the noisy datasets.

F. Model Robustness Analysis

We evaluate the robustness of our algorithms by introducing varying degrees of noise to the Houston 2013 dataset. In the

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experiment, we generate a noise matrix by randomly sampling from a standard normal distribution. Then, we adjust the noise level by multiplying the noise matrix with a scalar, denoted as μ, where $0 < \mu < 1$. For each image patch from different sources, a noisy dataset is generated by adding a randomly created noise matrix with a probability of 0.5. The classification results, as shown in Fig. 7, clearly indicate that while the performance of all the algorithms decreases with increased noise level, our algorithm consistently outperforms the others. These results suggest that even when the datasets contain noise, our algorithm can effectively suppress it and ensure promising classification performance. Despite the presence of noise, the unisource intrinsic information and cross-source relationship are still encoded into the multisource information bottleneck representation, resulting in robustness and superior generalization performance.

V. Conclusion

This article presents a novel multisource representation learning method termed MIBF-Net for multisource remote sensing image classification. The method integrates both unisource and cross-source information to comprehensively represent multisource remote sensing data, while simultaneously discarding redundant and noisy information to improve the discriminability of the resulting representation. As a result, the MIBF-Net method enhances the overall performance of multisource remote sensing classification. The experimental results demonstrate that the proposed method outperforms the existing methods, highlighting its effectiveness and generalizability. However, the proposed approach adopts a patch-based local learning framework, which achieves higher accuracy but also introduces a larger computational time overhead due to the overlapping image patches. In the near future, we plan to develop a patch-free global multisource remote sensing image classification framework that can improve the timeliness while maintaining high accuracy.

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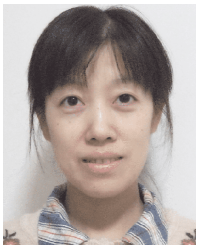
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